

HW 1 — Solutions, Worked in Full

Mathematical Foundations of Diffusion Models
Monsoon 2026 · IIT Jammu

Not a bare answer key: every part is derived, with the concept it is testing named

How to use this

These are full solutions to HW 1, written as derivations rather than as an answer key: each problem states the concept it is testing, works the computation, and names the error the problem was built to catch. Check your own script against them, and bring anything that still does not make sense to class.

The last problem is the one to reread. It builds — without saying so — exactly the object a diffusion model computes, and **Practice 1** picks it up from there and replaces its table with a Gaussian.

HW 1 ends by building a denoiser. Nobody told you that.

Problem 5 has a hidden cause Z , a noisy observation X , a prior, a likelihood, a posterior $\mathbb{P}(Z \mid X)$, and a conditional mean $\mathbb{E}[Z \mid X]$. That is, structurally and completely, what a diffusion model's denoiser computes. The only difference is that the homework's hidden cause takes two values and its observation takes three, so the whole thing fits in a 2×3 table.

Part A — HW 1, worked as derivations

1. Problem 1: a random variable is a function

The concept

$X : \Omega \rightarrow \mathbb{R}$ is an ordinary deterministic function. It is not random and it is not a variable. **All the randomness sits in which ω turns up**, and the function merely reports a number about it.

(a) $\Omega = \{1, 2, 3, 4, 5, 6\}$. An event is *any* subset, and a subset is built by making one independent in/out choice per outcome, so there are $2^6 = 64$ of them.

(b) $Y(\omega) = 1$ exactly when ω is prime:

$$\{\omega : Y(\omega) = 1\} = \{2, 3, 5\}, \quad \{\omega : Y(\omega) = 0\} = \{1, 4, 6\}.$$

(c) This is the part worth the time. The claim “ Y maps the event $\{2, 3, 5\}$ to 1” mistakes what Y eats. Its argument is a **single outcome**, never a set:

$$Y(2) = 1, \quad Y(3) = 1, \quad Y(5) = 1$$

— three separate evaluations that happen to return the same number. Nothing is ever evaluated at $\{2, 3, 5\}$. The set arises the other way round, as the **preimage** $Y^{-1}(\{1\})$: it is the *answer* to “where does Y return 1?”, not the *input*. The chain that is actually running:

outcome $\omega \xrightarrow{X}$ number $\longrightarrow$ condition $\longrightarrow$ event (a preimage) $\longrightarrow$ probability

(d) Name the event, then compute.

$$\mathbb{P}(Y = 1) = \mathbb{P}(\{2, 3, 5\}) = \frac{3}{6} = \frac{1}{2} = 0.500$$

$$\{X \leq 3\} \cap \{Y = 1\} = \{1, 2, 3\} \cap \{2, 3, 5\} = \{2, 3\}, \quad \mathbb{P} = \frac{2}{6} = \frac{1}{3} \approx 0.333$$

2. Problem 2: conditioning is division by a margin

The concept

Joint, marginal and conditional are three readings of **one** table. The joint is a cell over the net; a marginal sums a row or column; a conditional is a cell over its **row or column total**. Which total you divide by *is* the choice of what you are conditioning on.

Row totals 70, 70, 60; column totals 65, 70, 65; net 200. Both margins total 200 — your own consistency check.

(a) Divide by the net, because we want each group’s share of everything:

$$\mathbb{P}(X = \cdot) = \frac{1}{200}(70, 70, 60) = (0.350, 0.350, 0.300), \quad \mathbb{P}(Y = \cdot) = \frac{1}{200}(65, 70, 65) = (0.325, 0.350, 0.325).$$

(b) Now divide by the **row** total, because conditioning on $X = A$ restricts attention to that row and asks how it is distributed within itself:

$$\mathbb{P}(Y = \cdot \mid X = A) = \frac{1}{70}(40, 20, 10) = \left(\frac{4}{7}, \frac{2}{7}, \frac{1}{7}\right) \approx (0.571, 0.286, 0.143),$$

$$\mathbb{P}(Y = \cdot \mid X = C) = \frac{1}{60}(10, 15, 35) = \left(\frac{1}{6}, \frac{1}{4}, \frac{7}{12}\right) \approx (0.167, 0.250, 0.583).$$

Both sum to 1 — always check. The two rows are drastically different: grade 1 is the most likely outcome from machine A and the *least* likely from machine C . Knowing the machine changes what you expect of the grade, so X and Y are **dependent**.

(c) Conditioning on $Y = 2$ restricts to that **column**, so divide by its total:

$$\mathbb{P}(X = B \mid Y = 2) = \frac{35}{70} = \frac{1}{2} = 0.500.$$

Dividing by 200 instead gives $35/200 = 0.175$, which is the *joint* $\mathbb{P}(X=B, Y=2)$ — a different question, correctly answered.

(d) Under independence the expected count in a cell is row $\times$ column/net:

$$\frac{70 \times 65}{200} = 22.75 \quad \text{against an observed 40.}$$

Far more grade-1 parts come from A than independence would produce, so independence is untenable. Note carefully: agreement in *one* cell would not have established independence — that requires $\mathbb{P}(X=x, Y=y) = \mathbb{P}(X=x)\mathbb{P}(Y=y)$ in **every** cell.

3. Problem 3: Bayes, and why a good test gives a weak posterior

The concept

The posterior weighs the likelihood against the **base rate**. When the base rate is small enough, a large likelihood ratio is still not enough — because the false positives are drawn from a vastly larger pool.

(a) $\mathbb{P}(D) = 0.01$ (prior, unconditional); $\mathbb{P}(+ | D) = 0.95$ (sensitivity); $\mathbb{P}(+ | D^c) = 0.10$ (false-positive rate). The last two are conditionals *pointing in the opposite direction* to the one we want, which is the whole difficulty.

(b) Law of total probability over the partition $\{D, D^c\}$, disjoint and exhaustive by construction:

$$\mathbb{P}(+) = \mathbb{P}(+ | D)\mathbb{P}(D) + \mathbb{P}(+ | D^c)\mathbb{P}(D^c) = (0.95)(0.01) + (0.10)(0.99) = 0.0095 + 0.0990 = 0.1085.$$

Look at the two terms before moving on: 0.0095 from sick students, 0.0990 from healthy ones. **Ten times as many positives come from the healthy group**, and that ratio is the answer to (d) already sitting on the page.

(c) Bayes:

$$\mathbb{P}(D | +) = \frac{\mathbb{P}(+ | D) \mathbb{P}(D)}{\mathbb{P}(+)} = \frac{0.0095}{0.1085} = \frac{19}{217} \approx 0.0876.$$

(d) The condition is rare. In 10,000 screened students only 100 have it, of whom 95 test positive — but 990 of the 9,900 healthy students also test positive. True positives are outnumbered roughly 10:1.

Which number to change: the **false-positive rate**, downward. It is what generates the 990, and it dominates the denominator.

change	$\mathbb{P}(D +)$
as given	0.0876
sensitivity $0.95 \rightarrow 1.00$ (the best possible)	0.0917
false-positive rate $0.10 \rightarrow 0.05$	$95/(95+495) \approx 0.161$

Perfecting the sensitivity buys almost nothing; halving the false-positive rate nearly doubles the posterior.

The natural-frequency check, which makes the result obvious rather than merely correct:

	tests +	tests −	total
has condition	95	5	100
does not	990	8,910	9,900
total	1,085	8,915	10,000

$\mathbb{P}(D | +) = 95/1085 = 19/217$, identical. And note the table has turned Problem 3 into Problem 2 — same object, different presentation.

4. Problem 4: the continuous analogue

The concept

Everything in Problem 2 survives the move to densities with sums replaced by integrals. Marginalise = integrate the other variable out. Condition = divide by the margin. The normalising constant is the analogue of the net total.

(a) The constant makes the density integrate to 1 over its region — and the region is $[0, 1] \times [0, 2]$, not the unit square:

$$\int_0^2 \int_0^1 (x + y) dx dy = \int_0^2 \left(\frac{1}{2} + y \right) dy = \left[\frac{y}{2} + \frac{y^2}{2} \right]_0^2 = 1 + 2 = 3 \implies c = \frac{1}{3}.$$

(b) Integrate out the *other* variable, and carry the ranges:

$$f_X(x) = \int_0^2 \frac{1}{3}(x + y) dy = \frac{1}{3}(2x + 2) = \frac{2}{3}(x + 1), \quad 0 \leq x \leq 1,$$

$$f_Y(y) = \int_0^1 \frac{1}{3}(x + y) dx = \frac{1}{3} \left(\frac{1}{2} + y \right), \quad 0 \leq y \leq 2.$$

Checks: $\int_0^1 \frac{2}{3}(x + 1) dx = \frac{2}{3} \cdot \frac{3}{2} = 1$ and $\int_0^2 \frac{1}{3} \left(\frac{1}{2} + y \right) dy = \frac{1}{3}(1 + 2) = 1$.

(c) Divide the joint by the margin:

$$f(y | x) = \frac{f(x, y)}{f_X(x)} = \frac{\frac{1}{3}(x + y)}{\frac{2}{3}(x + 1)} = \frac{x + y}{2(x + 1)}, \quad 0 \leq y \leq 2.$$

Check: $\int_0^2 \frac{x + y}{2(x + 1)} dy = \frac{2x + 2}{2(x + 1)} = 1$ for **every** x — as it must be, since a conditional density is a genuine density in y for each fixed x .

What was divided by: $f_X(x)$ is the total mass along the vertical slice at that x . This is exactly dividing a row of a count table by its row total, and the whole of Problem 2(b) is the discrete shadow of this line.

(d) Not independent. Compare explicitly:

$$f_X(x) f_Y(y) - f(x, y) = \frac{2}{9}(x + 1) \left(\frac{1}{2} + y \right) - \frac{1}{3}(x + y) = \frac{1}{9}(y - 1)(2x - 1),$$

non-zero except on the lines $x = \frac{1}{2}$ and $y = 1$. Equivalently and faster: $x + y$ does not factor as (a function of x) $\times$ (a function of y), and a joint density factorises **if and only if** the variables are independent.

5. Problem 5: a hidden cause — the discrete denoiser

The concept

The whole tutorial in miniature. A cause Z you cannot see, an effect X you can, and the machinery that recovers a belief about the first from the second:

$$\text{prior} \times \text{likelihood} \rightarrow \text{joint} \rightarrow \text{marginal} \rightarrow \text{posterior} \rightarrow \text{conditional mean}$$

Problem 2 gave you the joint table for free because both variables were recorded. Here Z is **never observed**, so there is no table to build — only a model, a prior, and Bayes.

(a) Build the joint from $p(z, x) = p(z) p(x | z)$: multiply each row of the likelihood table by its prior.

	$X = 0$	$X = 1$	$X = 2$	row
$Z = 1$ (prior 0.6)	0.42	0.12	0.06	0.60
$Z = 2$ (prior 0.4)	0.08	0.12	0.20	0.40
column	0.50	0.24	0.26	1.00

(b) Sum out Z — the variable you do not observe:

$$\mathbb{P}(X = \cdot) = (0.50, 0.24, 0.26).$$

This is what the inspector actually sees, with the machine identity averaged away. Notice what has been destroyed: many different machine-mixtures produce this same marginal, so from $p(x)$ alone the split cannot be recovered. **Marginalising is lossy, and the lost thing is the decomposition.**

(c) Bayes, with the column of the joint table already serving as the numerator:

$$\mathbb{P}(Z = 1 \mid X = x) = \frac{\overbrace{\mathbb{P}(X = x \mid Z = 1)}^{\text{likelihood}} \overbrace{\mathbb{P}(Z = 1)}^{\text{prior}}}{\underbrace{\mathbb{P}(X = x)}_{\text{evidence}}}$$

x	computation	$\mathbb{P}(Z = 1 \mid X = x)$
0	0.42/0.50	21/25 = 0.840
1	0.12/0.24	1/2 = 0.500
2	0.06/0.26	3/13 $\approx$ 0.231

Read the trend: zero defects is evidence for the older machine, two defects is evidence against it, and one defect is exactly uninformative — it moves the belief from the prior 0.6 to 0.5, which is a *change*, so even the “uninformative” reading carries information.

(d) Z takes values in $\{1, 2\}$, so with $p = \mathbb{P}(Z = 1 \mid X = x)$ the average is taken *over the posterior of Z at that fixed x* :

$$\mathbb{E}_{Z \mid X=x}[Z] = 1 \cdot p + 2 \cdot (1 - p) = 2 - p.$$

x	$\mathbb{E}_{Z \mid X=x}[Z]$
0	29/25 = 1.160
1	3/2 = 1.500
2	23/13 $\approx$ 1.769

These are not in $\{1, 2\}$ and need not be. **A conditional expectation is a weighted average of the possible values, not a choice among them.** 1.16 says “probably machine 1, with some chance of machine 2” — it summarises a distribution rather than naming a label. If forced to name one, report the posterior *mode*, not the mean.

(e) The tower property, verified. Two *different* averages are involved, and the subscripts say which is which: the inner one runs over Z with x fixed (that is column (d)), and the outer one runs over X , weighting by how often each x occurs:

$$\mathbb{E}_X \left[\mathbb{E}_{Z \mid X}[Z \mid X] \right] = \underbrace{(0.50)}_{\mathbb{P}(X=0)}(1.160) + \underbrace{(0.24)}_{\mathbb{P}(X=1)}(1.500) + \underbrace{(0.26)}_{\mathbb{P}(X=2)}(1.769) = 0.58 + 0.36 + 0.46 = 1.40,$$

against $\mathbb{E}_Z[Z] = 1(0.6) + 2(0.4) = 1.40$. Equal exactly, not approximately.

Note what the inner object is: $\mathbb{E}_{Z|X}[Z | X]$ is not one number but a **random variable**, taking the values 1.160, 1.500, 1.769 with probabilities 0.50, 0.24, 0.26. The outer $\mathbb{E}_X$ is what averages it. §7.0 returns to this distinction, which is the one that makes the tower property look circular when the subscripts are dropped.

In words: averaging the conditional averages, **weighted by how often each condition occurs**, returns the overall average. Observing X redistributes your estimate of Z but cannot move it on average — the precise sense in which an observation is informative about *cases* without being biased overall.

Where this goes next

Computation	Where it came from	Where it returns
Prior $\times$ likelihood $\rightarrow$ posterior	HW 1 P3, P5	the reverse process; $q(x_{t-1} x_t, x_0)$
Marginalising a latent	HW 1 P5(b)	the intractable $p_\theta(x_0)$; why the ELBO exists
Tower property	HW 1 P5(e)	every expectation argument in Module 3
$\mathbb{E}_{Z X}[Z X]$ as a weighted average	HW 1 P5(d)	the denoiser output

The concept

The one sentence. Problem 5 is a diffusion model’s denoiser in miniature: a hidden clean quantity, a noisy observation, a posterior over the hidden quantity, and its mean.

A real diffusion model runs that same pipeline at every noise level at once, with the 2×3 table replaced by a Gaussian — at which point the posterior becomes a logistic function and the conditional mean becomes a tanh. That derivation is **Practice 1**, which starts exactly where this leaves off.